

Explosion Dashboard

Development Design

ID 64004

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Explosion Dashboard

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Explosion Dashboard

Introduction

Create a new Explosion dashboard which will consolidate pertinent information together allowing users to research a product without having to access multiple pages.

Objective

Automate nearly all the manual steps in the current process by bringing together the most pertinent information [raw numbers and calculated results] onto an Explosion Dashboard.

The page will offer multiple links to existing pages some of which will be enhanced so the user will be only one click away from information that used to require them to visit multiple pages.

The page will offer columns which display the most important, at-a-glance information needed by the user and provide an option to create a CSV download file that will allow to take all of the information to a spreadsheet for further analysis.

Explosion Dashboard

Scope

This project will be designed and delivered with the following release and version levels:

HarrisData HD5.0 115166
IBM i V7R1M0

This project will be based on the HTML User Interface. The Legacy User Interface will continue to work as it does today.

Minor changes may be made to the design as the enhancement is coded. These changes are made to correct issues discovered in developing the enhancement that were not found during the design. If any changes are made they will be included in the copy of the design you receive with the final product delivery. This design will also serve as your documentation until the User Documentation is updated.

Explosion Dashboard

Programming Requirements

Explosion Dashboard

When the new 'Explosion Dashboard' is selected from the Item/Plant Select page (Figure 3), the Indented Explosion Dashboard (Figure 1) will be displayed and provide the following functionality:

- Header icons
 - The dashboard icon  links to the Where Used Dashboard for the Item/Plant.
 - The display bar graph  and hide bar graph  icons toggle the display of the bar graphs shown in Figure x.
 - The home icon  will return the user to the Item/Plant Select page (Figure 3).
 - The download icon  will generate a csv file and place it in your current download folder. Refer to the Download CSV File section for additional information.
 - The printer friendly icon  will open a new tab and display the current page excluding the menu bar.
 - The help icon  will display this design in a new page tab.
- Header Information displayed for the selected Parent Item:
 - Plant – links to the Plant Select page
 - Item – links to the Item/Plant Select page (same as the home icon)
 - Quantity On Hand – links to Stock Variance
 - On Hand Floorstock
 - Quantity Available – links to Available To Promise
 - Cost (if displayed and Part Type is 'M' or 'B') – links to Cost Simulator

Explosion Dashboard

- Anticipated Annual Usage – link will default the Anticipated Annual Usage quantity to the Quantity Required and set the Usage From to the beginning of the current year (0101yy where yy is the current year) and Usage To will be set to the current date.
- Pegged Requirements
- Usage Selection
 - Usage From/To Dates are used to fill the following quantities in the work file (PDM099). If no dates entered, Total Usage and Percent of Total cannot be calculated.
 - Quantity Sold (PZQSLD)
 - Sum of Transaction Quantity (DTQTY) in Transaction History (HDDTRN) for Transaction Type SLOE where Transaction Date (DTTRDT) between Usage From/To
 - Quantity Issued (PZQISS)
 - Sum of Transaction Quantity (DTQTY) in Transaction History (HDDTRN) for Transaction Types ('ISIN','IFIN','KIOU') where Transaction Date (DTTRDT) between Usage From/To
 - The stop icon  will clear the Usage From/To dates and reset the Quantity Required to 1.
- Quantity Required
 - If blank and Usage From/To are entered, default to:
 - Sum of Transaction Quantity (DTQTY) in Transaction History (HDDTRN) for Transaction Types ('SLOE','ISIN','IFIN','KIOU') where Item Number (DTITEM) = Parent Item and Plant (DTPLT) = Plant and Transaction Date (DTTRDT) between Usage From/To
 - If blank and Usage From/To are blank, defaults to 1
 - Value can be overridden
 - This value (defaulted or entered) is used to calculate the extended Quantity Required for each Component based on the Quantity Per.

Explosion Dashboard

- View checkboxes:
 - Level 1 - if checked, only rows with Level = 1 are displayed
 - With Total Usage – if checked, only rows with Total Usage > 0 are displayed
- Column Links
 - Level – links to the Where Used Dashboard
 - Component – links to Item Select
 - Description – links to Item/Plant Select
 - Quantity On Hand – links to Stock Variance
 - Quantity Available – links to Available To Promise
 - Cost (if displayed and Part Type is 'M' or 'B') – links to Cost Simulator
- Hover Text on the following columns will display the calculation of the value displayed. The examples shown are for Component M10 (Figure 1).
 - Total Usage
 - Sold + Issued ($842.0000 = 503.0000 + 339.0000$)
 - Percent of Total
 - Required / Total Usage ($.071 = 60.0000 / 842.0000$)
 - Extended Cost (if displayed)
 - Quantity On Hand * Cost ($106136.00 = 13267.0000 * 8.0000$)

Explosion Dashboard

Explosion Dashboard

Plant: Universal Manuf. MKE [11]
Item: Flying 18" [101505]

Quantity On Hand: 5,224.0000
On Hand Floorstock: 87.0000
Quantity Available: 2,686.0000

Cost/Extended: 28.4311 / 148,524.1708
Anticipated Annual Usage: 726.0000
Pegged Requirements: 2,476.0000

Usage From	Usage To	Quantity Required
010119	123119	20

Page: 1 Search: Component View: ☐ Level 1 ☐ With Total Usage

Level	Routing Seq	Component	Description	Stock UOM	Part Type	Quantity Per	Quantity Required	Total Usage	Percent of Total	Quantity On Hand	Quantity Available	Cost	Extended Cost
1	10	EGGS	Incredible/edible (single) egg	EA	P	1.0000	20.0000	0.0000	0.00 %	158.9996	(226.0004)	1.0000	158.99
1	10	GLUE	Glue	OZ	P	0.3300	6.6000	0.0000	0.00 %	5,014.2100	4,738.0122	0.2000	1,002.84
1	10	M20	Armature Spring	EA	M	1.0000	20.0000	12.0000	166.66 %	197.0000	(614.6600)	25.2753	4,979.24
.2	10	M10	Test item	EA	M	3.0000	60.0000	842.0000	7.12 %	13,266.0000	12,000.0000	8.0000	106,128.00
.3	10	308540	Steel Tubing 1/2" X 10'	FT	P	1.0000	60.0000	511.5000	11.73 %	51,597.8750	(5,702.0000)	1.0000	51,597.87
.3	10	308805	1/8" Bar Stock	EA	P	1.0000	60.0000	12,560.0000	0.47 %	17,873.0000	(894,293.0000)	1.0000	17,873.00
1	666	P31	Bolt, Hex Head, 3/4"	EA	P	1.0000	20.0000	13,923.0000	0.14 %	30,581.0000	29,157.3400	1.0000	30,581.00

Figure 1

Explosion Dashboard

The bar graphs display YTD Units for the last three years for Sales, Issues and Total for the Parent Item.

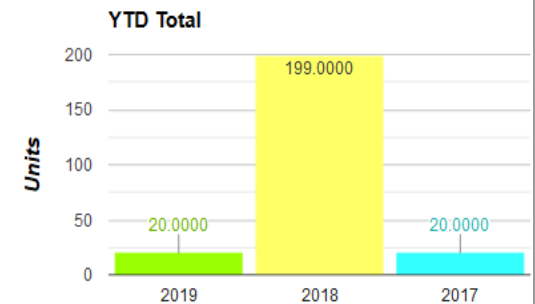
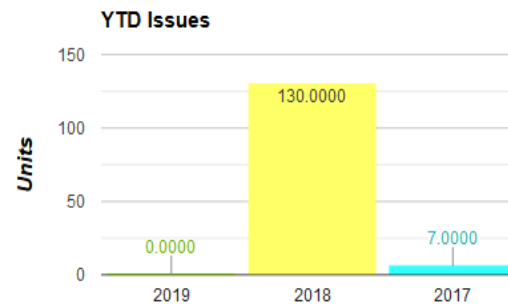
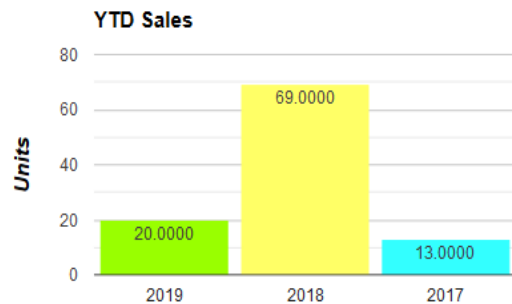
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Page: 1 Search: Component View: ☐ Level 1 ☐ With Total Usage

Level	Routing Seq	Component	Description	Stock UOM	Part Type	Quantity Per	Quantity Required	Total Usage	Percent of Total	Quantity On Hand	Quantity Available	Cost	Extended Cost
1	10	EGGS	Incredible/edible (single) egg	EA	P	1.0000	20.0000	0.0000	0.00 %	158.9996	(226.0004)	1.0000	158.99
1	10	GLUE	Glue	OZ	P	0.3300	6.6000	0.0000	0.00 %	5,014.2100	4,738.0122	0.2000	1,002.84
1	10	M20	Armature Spring	EA	M	1.0000	20.0000	12.0000	166.66 %	197.0000	(614.6600)	25.2753	4,979.24
2	10	M10	Test item	EA	M	3.0000	60.0000	842.0000	7.12 %	13,266.0000	12,000.0000	8.0000	106,128.00
..3	10	308540	Steel Tubing 1/2" X 10'	FT	P	1.0000	60.0000	511.5000	11.73 %	51,597.8750	(5,702.0000)	1.0000	51,597.87
..3	10	308805	1/8" Bar Stock	EA	P	1.0000	60.0000	12,560.0000	0.47 %	17,873.0000	(894,293.0000)	1.0000	17,873.00
1	666	P31	Bolt, Hex Head, 3/4"	EA	P	1.0000	20.0000	13,923.0000	0.14 %	30,581.0000	29,157.3400	1.0000	30,581.00

Figure 2

Explosion Dashboard

Item/Plant Select

Add the following quick links to the Item/Plant Select page (Figure 3). All warehouses that supply the plant will be included.

- Customer Order History
- YTD Top Customers
- YTD Top Vendors
- Also add a quick link option for [Explosion Dashboard](#) which links to (Figure 1).

Item/Plant Select

Plant: Universal Manuf. MKE [11]

Item: 1/8" Bar Stock [308805]

Part Type: Purchased [P]

Attachments

Available To Promise

Cost Summary

Current Quantities

Customer Order History

Demand

Demographics

Explosion Dashboard

Usage By Period

Usage By Year

Vendor Open Orders

Vendor Order History

Warehouses

Where Used Dashboard

Whs Usage By Period

Whs Usage By Year

YTD Top Customers

YTD Top Vendors

Customer Order History (5 Of 17 Rows)

Customer Number	Customer Name	Order Type	Order Number	Line	Rel	Required Date	Quantity Ordered	Quantity Shipped	Price	Extended Price
90	Rapid Air M5	O	6401	1	0	09-12-2019	1,500.0000	750.0000	10.00	7,500.00
90	Rapid Air M5	O	6401	1	0	09-12-2019	1,500.0000	750.0000	10.00	7,500.00
90	Rapid Air M5	O	6207	1	0	06-14-2018	155.0000	155.0000	10.00	1,550.00
90	Rapid Air M5	O	6109	15	0	02-07-2018	7.0000	7.0000	10.00	70.00
95	Rapid Air - Bill To	O	6110	15	0	02-07-2018	7.0000	7.0000	10.00	70.00

YTD Top Customers (1 Of 1 Rows)

Total Sales: \$15,000.00

Customer Number	Customer Name	Sales	Percent of Total	Quantity
90	Rapid Air M5	15,000.00	100.00 %	1,500.0000

YTD Top Vendors (1 Of 1 Rows)

Total Purchases: \$30,000.00

Vendor Number	Vendor Name	Purchases	Percent of Total	Quantity
129	Williams Steel Co	30,000.00	100.00 %	30,000.0000

Figure 3

Explosion Dashboard

Explosion Dashboard

Download CSV File

When the download icon  is selected from the Indented Explosion Dashboard (Figure 1), a csv file is created in your current download folder with the name: Indented_Explosion_<plant>_<item>.csv (i.e. Indented_Explosion_11_101505.csv) and will contain all of the columns in the Indented Product Structure work file (PDM099) (except for Hex Handle and costs if not authorized) and will also include the header information of the Parent Item (which repeats for each Component) as shown in the following example:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Row Seq	Plant	Parent Item	Parent Desc	Parent Cost	Parent Extended Cost	Parent Qty On Hand	Parent AAU	Parent Qty Avail	Parent Qty Floorstock	Parent Pegged Req	From Date	To Date	Quantity Required
2	1	11	101505	Flighting 18"	29.0911	151,972.01	5,224.00	726	2,686.00	87	2,421.00	1/1/2019	12/31/2019	20
3	2	11	101505	Flighting 18"	29.0911	151,972.01	5,224.00	726	2,686.00	87	2,421.00	1/1/2019	12/31/2019	20
4	3	11	101505	Flighting 18"	29.0911	151,972.01	5,224.00	726	2,686.00	87	2,421.00	1/1/2019	12/31/2019	20
5	4	11	101505	Flighting 18"	29.0911	151,972.01	5,224.00	726	2,686.00	87	2,421.00	1/1/2019	12/31/2019	20
6	5	11	101505	Flighting 18"	29.0911	151,972.01	5,224.00	726	2,686.00	87	2,421.00	1/1/2019	12/31/2019	20
7	6	11	101505	Flighting 18"	29.0911	151,972.01	5,224.00	726	2,686.00	87	2,421.00	1/1/2019	12/31/2019	20
8	7	11	101505	Flighting 18"	29.0911	151,972.01	5,224.00	726	2,686.00	87	2,421.00	1/1/2019	12/31/2019	20

	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB
1	Level	Routing Seq	Component Item	Component Desc	Stock UOM	Part Type	Backflush	Order Policy	Lead Time Days	Lead Time Code	Quantity Per	Ext Quantity Per	OnHand Quantity	OnHand Floorstock
2	1	10	EGGS	Incredible/edible (single) egg	EA	P		A	0 F		1	20	158.9996	158.9996
3	1	10	GLUE	Glue	OZ	P		P	15 F		0.33	6.6	5014.21	5014.21
4	1	10	M20	Armature Spring	EA	M	S	A	10 F		1	20	197	197
5	0.2	10	M10	Test item	EA	M		P	16 V		3	60	13266	13233
6	..3	10	308540	Steel Tubing 1/2" X 10'	FT	P		P	0 F		1	60	51597.875	40317.875
7	..3	10	308805	1/8" Bar Stock	EA	P	F	P	10 F		1	60	17884	17551
8	1	666	P31	Bolt, Hex Head, 3/4"	EA	P	S	A	0 F		1	20	30581	-10494

Explosion Dashboard

	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM
1	OnHand Receiving	Qty Held In Stock	Quantity On Order	Qty Committed Mfg	Available Quantity	Quantity Sold	Quantity Issued	Total Usage	Percent of Total	Cost	Extended Cost
2	0	0	9.0004	385	-226.0004	0	0	0	0	1	158.99
3	0	0	0	276.1978	4738.0122	0	0	0	0	0.2	1002.84
4	0	0	526	811.66	-614.66	0	0	0	0	25.27535	4979.24
5	33	0	10608	1266	12000	0	0	0	0	8	106128
6	11280	0	0	46019.875	-5702	0	0	0	0	1	51597.87
7	333	0	110	912166	-894293	0	0	0	0	1	17873
8	41075	0	7873	1423.66	29157.34	0	0	0	0	1	30581

	AN	AO	AP	AQ	AR	AS	AT	AU
1	Component Disposition	Inventory Use-Up Date	Accounting Lot Size	Fixed Order Quantity	Minimum Order Quantity	Maximum Order Quantity	Multiple Order Quantity	Safety Stock Quantity
2	I	11/3/2019	1	0	0	0	0	0
3			100	0	0	0	100	0
4			100	0	0	0	0	0
5			1	0	0	0	0	0
6			1	0	0	0	1	50
7			100	0	0	0	100	85
8			1	0	0	0	0	0

Explosion Dashboard

Program Option Security

A new 'ITEMCOST' program name will be added to Program Option Security (Figure 4) which will control the display of costs on the new dashboards. This includes the Parent Item Cost in the header and the Cost and Extended Cost columns for each Component (Figure 1). To easily restrict all users from viewing cost, add the 'HDS' User Profile with Display Cost not selected. Then add any specific User Profiles that should view cost with the Display Cost flag selected.

Program Option Security Maintenance

Program Name: [ITEMCOST]

User Profile	Name	Role

Page: 1 of 1

Opt	User Profile	Name	Role	Display Cost
	HDS	Program Option Security	HD_SECURE	☐
	RDBHRPR	Brian	HD_ALL	☒
	SPMCMPR	Matt, you handsome devil!	HD_ALL	☒

Figure 4

Explosion Dashboard

Database Requirements

Indented Product Structure Work (PDM099)

Column	Description	Start	End	Bytes	Dc	T
PZXHND	Hex Handle	1	64	64	A	
PZRSEQ	Row Sequence	65	69	5	0	S
PZPLT	Plant Number	70	72	3	0	S
PZPN	Item Number	73	87	15	A	
PZFDAT	From Date	88	94	7	0	S
PZTDAT	To Date	95	101	7	0	S
PZQREQ	Quantity Required	102	108	13	4	P
PZLVL	BOM Level	109	116	8	A	
PZSEQN	Routing Sequence Number	117	119	3	0	S
PZCPN	Component Item Number	120	134	15	A	
PZIMDS	Item Description	135	164	30	A	
PZUOMS	Stocking Unit Of Measure	165	166	2	A	
PZPTYP	Part Type (M,P,B,S)	167	167	1	A	
PZSTWC	Backflush Code	168	168	1	A	
PZOPC	Order Policy Code	169	169	1	A	
PZLTDY	Lead Time Days	170	173	4	0	S
PZLTC	Lead Time Code	174	174	1	A	
PZQPER	Quantity Per	175	179	9	5	P
PZXPER	Extended Quantity Per	180	186	13	4	P
PZOHQT	On Hand Quantity	187	193	13	4	P
PZOHFS	On Hand Floor Stock	194	200	13	4	P
PZOHRC	On Hand Receiving	201	207	13	4	P
PZQHSR	Qty Held In Stockroom	208	214	13	4	P
PZQOO	Quantity On Order	215	221	13	4	P
PZCMTO	Qty Committed To MFG	222	228	13	4	P
PZAVQT	Available Quantity	229	235	13	4	P
PZQSLD	Quantity Sold	236	242	13	4	P
PZQISS	Quantity Issued	243	249	13	4	P
PZQTOT	Total Usage	250	256	13	4	P
PZQPCT	Percent of Total	257	262	11	6	P
PZTUCC	Cost	263	268	11	5	P
PZEXTC	Extended Cost	269	274	11	2	P
PZECDP	ECN Component Disposition (b,I)	275	275	1	A	
PZIUPD	ECN Inventory Use-Up Date	276	285	10	0	L
PZALTS	Accounting Lot Size	286	291	11	0	P
PZFOQ	Fixed Order Quantity	292	297	11	0	P
PZMNQ	Minimum Order Quantity	298	303	11	0	P
PZMXQ	Maximum Order Quantity	304	309	11	0	P
PZMUQ	Multiple Order Quantity	310	315	11	0	P
PZSSQ	Safety Stock Quantity	316	321	11	0	P